

# Developer Guidelines for Automated Pile Load Test Report Generation

## Target Audience and Input Constraints

These guidelines are intended for a developer or coding agent responsible for programmatically generating pile load test reports from **image-based data inputs**. The agent will be extracting structured data from screenshots or scanned images (e.g. tables of test results, graphs of load vs displacement) rather than parsing PDFs or PPT files directly. Therefore, the focus is on robust **OCR-based data extraction** and interpretation of visual content. The goal is to produce a well-structured, comprehensive report in a consistent format, given the challenges of imperfect OCR and image inputs. The coding agent should be prepared to apply domain knowledge and use AI assistance where needed to interpret the data (especially graphs or unclear text), as detailed below.

## Test Types Covered

The system must handle report generation for the following pile load test types, each with distinct procedures and acceptance criteria:

- **Initial Vertical Pile Load Test (IVPLT):** An intensive vertical load test on a **test pile** (often non-working pile) to **2.5× the design load** <sup>1</sup>. It evaluates ultimate capacity with stricter settlement criteria (typically 12 mm limit).
- **Routine Vertical Pile Load Test (RVPLT):** A routine proof test on a **working pile** to **1.5× the design load** <sup>2</sup>. It confirms working load capacity with a slightly higher settlement tolerance (typically 18 mm limit).
- **Lateral Load Test:** A horizontal load test (often initial) to check lateral capacity. Often conducted to **2.5× the design lateral load** (if initial), usually under a **5 mm lateral deflection** acceptance criterion. It uses a reaction frame or pile to apply sideways load.
- **Uplift/Pullout Test:** A vertical uplift (tension) test (usually initial) to **2.5× the design uplift load**. It checks anchorage capacity; acceptance often based on a 12 mm upward movement limit or a yield point in the load-displacement curve.

Each test type will follow the general report structure outlined below, with some differences in content details, criteria, and phrasing as noted.

## Report Structure and Mandatory Sections

Each generated report must contain the following primary sections (in this order) to mirror industry-standard pile test reports <sup>3</sup> <sup>4</sup>:

- **1.0 General:** Introduction describing the test's context and purpose.

- **2.0 Scope of Work:** Details of the pile tested and test parameters.
- **3.0 Methodology:** Description of how the test was conducted, referencing standards.
- **4.0 Results:** Key findings, including acceptance criteria and whether they were met.
- **5.0 Readings and Graphs:** Presentation of detailed field readings (tables) and plotted load vs. displacement graphs.
- **(Optional) Field Readings / Data:** Raw data tables (if not included in section 5.0) and any observations during testing.
- **(Optional) Calibration Certificate:** If available, a section appending calibration certificates for instruments.

Below, we detail the content and language expected in each section for each test type.

## Initial Vertical Load Test (IVPLT)

**General:** State the project and purpose. For example: *“Clients decided to carry out a static vertical pile load test on a 600 mm diameter pile to estimate its load carrying capacity and settlement. <sup>5</sup> M/s <Agency> was entrusted to perform this initial test.”* Mention relevant codes (e.g. IS 2911 (Part 4):2013 for load tests <sup>6</sup> ).

**Scope of Work:** Summarize pile details in a table or text: location, pile ID, diameter, depth, concrete grade, etc. Include design load and the required test load. For an initial test, confirm that **test load = 2.5 × design load** (e.g. *design load 147 T, tested to 367.5 T <sup>1</sup>*). This section often explicitly states the test load requirement: *“The pile is required to be tested to a load of 367.5 T.”* Include a subsection “Pile details for Initial Pile” with these parameters.

**Methodology:** Describe the test setup and procedure, referring to IS codes. Note that for initial tests, loading is done in increments (typically ~20% of design load per step <sup>7</sup> ) up to 2.5× design load. Mention the use of hydraulic jacks, reaction systems, dial gauges, and the load application procedure. For example: *“The initial load test was carried out as per IS:2911 (Part 4). The pile was loaded in increments of 20% of the design load, up to the test load, and then unloaded as per the prescribed sequence <sup>7</sup> . A hydraulic jack of adequate capacity (with calibrated pressure gauge) applied the load, reacting against a sturdy reaction frame. Four dial gauges (0.01 mm least count) were fixed on the pile head to record settlements <sup>8</sup> <sup>9</sup> . Each load increment was held for 1 hour, and readings were taken at 1, 5, 10, 15, 30, and 60 minute intervals.”* Also mention if a **24-hour hold** was done at max load (common in initial tests for creep observation). Calibration verification should be noted: e.g. *“Calibration charts for the jack and gauges were checked before testing.”*

**Results:** Begin by stating the **Acceptance Criteria for vertical load** as per IS:2911. For initial tests, explicitly list the criteria (usually provided in bullet form in reports): for example: *“The Safe Capacity of the pile is considered as the lesser of: (a) two-thirds of the load at 12 mm total settlement (or 2% of pile diameter, 12 mm in this case) <sup>10</sup> ; (b) 50% of the load at 10% of pile diameter settlement (60 mm) <sup>11</sup> .”* Next, present the actual test outcomes against these criteria. Include the maximum settlement observed at peak load, the elastic rebound upon unloading, and the net settlement. For example: *“Maximum settlement at 382.65 T (after 24 hrs) = 9.88 mm; Total rebound = 2.36 mm; Net settlement = 7.52 mm <sup>12</sup> .”* State whether this meets the criteria (in this example, 7.52 mm net < 12 mm allowable). Conclude with a clear statement of capacity: a suggested wording pattern is: **“As per the test data and graph, the test pile has shown greater load-carrying capacity than the design load of X T. Therefore, X T is adopted as the safe vertical load for the working piles.”** In our example: *“...the test pile has shown more load carrying capacity than the design load of 147 T <sup>13</sup> . So, 147 T can be adopted as the safe vertical load for the working piles <sup>14</sup> .”* This active-voice

conclusion (“So, X T can be adopted...”) should be included for all passed tests. If the criteria were **not** met (e.g. excessive settlement), use a different phrasing (see **Edge Cases** below).

**Readings and Graphs:** Provide the load-settlement data and graph. This typically includes:

- **Tables of Load vs Settlement:** One table for the **loading cycle** (incremental load and corresponding settlements at time intervals), and sometimes a second for the **unloading cycle**. The table columns usually include Time, Pressure (kg/cm<sup>2</sup>), Calculated Load (T), and multiple dial gauge readings with an average settlement <sup>15</sup>. Ensure the table clearly distinguishes loading and unloading phases (you may separate them or include a “Remarks” column indicating “Loading”/“Unloading”). Also include a summary row for residual settlement after unloading if available.
- **Graph:** A plot of Load (on the y-axis) versus Settlement (on the x-axis) for the pile. Label the axes with units (e.g., “**Settlement (mm)**” and “**Load (T)**”), and indicate important points such as the design load and the defined safe load on the graph (e.g., horizontal/vertical lines or annotations). The graph should show the pile’s load-displacement curve; if unloading data is available, a loop curve can be included, but a primary loading curve is essential. Mark any defined **failure point or allowable limit** on the graph (e.g., a dashed line at 12 mm settlement). *Note:* If graph data must be derived from the table, ensure smoothing is done as needed and consider using an LLM to identify any “break point” in the curve (for uplift tests) as discussed later. The agent should output or reference the graph image in the report JSON (if the frontend will render it from data).

*(The Calibration Certificate section, if present, would follow, but since it is standard for all tests, it is described under Edge Cases and additional notes.)*

## Routine Vertical Load Test (RVPLT)

**General:** Similar to IVPLT, but clarify this is a **routine test on a working pile**. E.g., “*This report covers a routine vertical pile load test on a 1200 mm dia working pile to verify its safe load capacity* <sup>16</sup>.” Mention the project context and that it’s done according to IS:2911 as well.

**Scope of Work:** List pile details as for IVPLT. For routine tests, ensure **test load = 1.5 × design load** (e.g. design 550 T, test to 825 T <sup>17</sup> <sup>18</sup>). State the pile ID/location, diameter, depth, concrete grade, etc. and the safe capacity/design load values. For example: “*Safe capacity (design load) = 550 T; Test Load = 825 T; Pile diameter = 1200 mm; Depth = 36 m...*” <sup>17</sup>. If applicable, note this test was performed at working pile cutoff level.

**Methodology:** The procedure is largely the same as IVPLT, with possibly shorter hold times since it’s a smaller multiple of load. Describe loading in 20% design load increments <sup>7</sup>, use of one or more jacks, reaction system, etc. A routine test might not require a 24-hour hold at max load (often each load is held for 1 hour, including the maximum load, unless creep is suspected). State that “*The routine test was carried out to 1.5 times the design load as per IS:2911* <sup>2</sup>. *Each increment (≈20% of design load) was maintained for at least 1 hour. Settlements were recorded using dial gauges (0.01 mm LC) at four locations on the pile head.*” Emphasize any differences: e.g., if only one jack or one gauge is used (though typically 2–4 gauges are used even in routine tests for accuracy).

**Results:** State acceptance criteria for routine vertical tests. This is similar to initial tests but with a larger settlement limit: *“Safe load is the lesser of: (a) two-thirds of the load at 18 mm total settlement (or 2% of diameter, 24 mm for 1200 mm dia, whichever is less) <sup>19</sup>; (b) 50% of load at 10% diameter settlement (120 mm).”* (Note: IS:2911 allows 12 mm for initial tests and 18 mm for routine, hence the difference). Present the test results: *“Maximum settlement at 825 T = 5.79 mm, rebound = 1.92 mm, net settlement = 3.87 mm <sup>20</sup>.”* Clearly, this is well below 18 mm, so criteria are satisfied. Conclude with a similar line in active voice: *“Thus, the pile showed more capacity than the 550 T design load, and 550 T is adopted as the safe vertical load for this pile <sup>21</sup>.”* If the test data includes a 24-hour reading (less likely for routine), include it in the max settlement if provided.

**Readings and Graphs:** Provide the load-settlement table and graph as in IVPLT. The format is the same: time, pressure, load, multiple dial readings, etc. Because the test load is lower (e.g. 825 T in the example) and settlement small, ensure the graph's scale is appropriate to show the curve (you may mark the 18 mm criterion on the graph even if not reached, for context). The graph for routine tests often looks almost linear since working loads don't reach failure – that is normal. Still, label axes and annotate design load and safe load. If any creep or 24-h hold data was recorded (sometimes a 24-h unload rebound check is done), include that in the table/graph or note it.

## Lateral Load Test

**General:** State that a lateral (horizontal) load test was conducted, including context (free-head or fixed-head condition, usually free-head for field test). For example: *“Clients decided to carry out a lateral pile load test on a 600 mm dia pile to estimate its lateral load capacity and deflection <sup>22</sup>. This initial lateral load test was performed as per IS:2911 (Part 4) – 2013 on a free-head pile.”* Identify it as initial or routine if applicable (often lateral tests are done as initial tests on test piles, but it can be routine if required by the project).

**Scope of Work:** Tabulate details: pile ID and location, pile diameter and depth, design lateral load, and target test load. For initial lateral tests, target is typically 2.5× design lateral load. E.g., *“Design lateral load = 3.5 T, Test load = 8.75 T (2.5 × design) <sup>23</sup> <sup>24</sup>.”* Include pile concrete grade and any specific conditions (e.g., ground level, whether the test is at cutoff level or requires a certain embedment for reaction). If it's a routine lateral test (less common), the target might be 1.5× design.

**Methodology:** Describe the lateral load setup. For example: *“The test was conducted with a hydraulic jack of 50 T capacity reacting against a reaction pile/structure <sup>25</sup>. The jack was placed horizontally and push load was applied to the test pile (free-head condition). Two dial gauges (0.01 mm LC) were positioned on the test pile, 180° apart, to measure lateral deflection, and similarly on the reaction point to account for any movement of the reaction system <sup>26</sup> <sup>27</sup>. The gauges were set to zero at the start.”* Explain that load increments ~20% of estimated safe load were applied and held until rate of displacement was very slow (e.g., <0.1 mm per 30 min) <sup>28</sup> <sup>29</sup>. Often, each load is held for 5–15 minutes or until displacement stabilizes, with a longer hold at the maximum load (e.g., 30–60 min) <sup>30</sup> <sup>31</sup>. Note that the **loading and unloading sequence** should be described: *“Loading was done in increments of ~0.7 T (approximately 20% of 3.5 T design load) up to 8.75 T <sup>32</sup> <sup>33</sup>. Because the pressure gauge least count is 5 kg/cm<sup>2</sup>, the actual increments were rounded to the nearest achievable load <sup>34</sup>. Each increment was maintained for at least 30 min, and the final load was held for 60 min <sup>30</sup> before unloading. The pile's lateral displacement was monitored throughout.”* Mention that unloading was done similarly in steps, and residual deflection (permanent set) was noted if any. Calibration of jack and gauge should also be mentioned (usually done beforehand).

**Results:** State the acceptance criterion for lateral tests. Commonly, **the safe lateral load is defined as the load at 5 mm lateral deflection** for initial tests (or another specified deflection per project criteria). In the example documents, it is given as: *“Final load at which total displacement corresponds to 5 mm”* <sup>35</sup>. This essentially means the pile should not reach 5 mm deflection at the safe (design) load. Report the actual performance: list the maximum deflection observed at the test load, the rebound on unloading, and net permanent deflection. For example: *“Maximum lateral displacement at 8.90 T = 2.84 mm; Total rebound = 1.08 mm; Net deflection = 1.76 mm”* <sup>36</sup>. Since 2.84 mm < 5 mm, the criteria are satisfied in this case. Use a conclusion phrasing such as: *“From the load–deflection data and graph, the pile did not reach 5 mm deflection even at the full test load of 8.75 T. The pile has shown more lateral load carrying capacity than the design load of 3.5 T, with deflections within permissible limits up to the test load”* <sup>37</sup>. Therefore, 3.5 T can be adopted as the safe lateral load for the working piles <sup>37</sup>. This communicates that the design lateral capacity is confirmed. If the test had reached 5 mm at some lower load, that load would be used to determine safe capacity (e.g., two-thirds of that load if required by local practice, or simply that load as safe load).

**Readings and Graphs:** Include a table of lateral load test readings and a load–deflection graph:

- **Table:** The lateral test table should capture time or hold duration, pressure, load, and deflections of both the test pile and reaction point. Typically columns might be: Time (min), Pressure (kg/cm<sup>2</sup>), Calculated Load (T), Dial Gauge Readings at test pile (maybe two gauges) and at reaction (two gauges), and then the **average deflection** of the test pile and reaction pile <sup>38</sup> <sup>27</sup>. Often, the report computes net deflection by subtracting reaction movement from test pile movement. Ensure the agent averages the multiple readings correctly and computes net deflection = (avg. test pile deflection) – (avg. reaction deflection). Present data for each load increment and unload step, clearly indicating the sequence.
- **Graph:** Plot Load (T) vs Lateral Deflection (mm). Typically, deflection is on the x-axis (mm) and load on the y-axis (T) for easy reading of “load at 5 mm deflection.” Label axes and mark the 5 mm deflection line. Show the curve of load up to max test load. If the pile didn’t reach 5 mm, the curve will end before that vertical line, indicating additional capacity. If any permanent set (residual deflection) occurred, it can be indicated by an offset on the x-axis upon unloading (optional). Annotate the design load and perhaps the target 5 mm criterion on the graph.

*(No separate calibration section is usually needed beyond noting that gauges were calibrated, unless a certificate is appended.)*

## Uplift/Pullout Load Test

**General:** Outline that an uplift (tension) test was performed on a pile, including purpose and standards. E.g., *“An initial uplift (pullout) pile load test was conducted on a 600 mm dia pile to determine its uplift capacity”* <sup>39</sup>. This test was carried out as per IS:2911 (Part 4):2013 for tension load on piles.” Highlight that it’s an uplift test (behavior is somewhat different from compression).

**Scope of Work:** Provide pile details similar to vertical tests: pile ID, location, diameter, depth, concrete grade, design uplift (safe) load, and target test load. For initial uplift tests, target is typically **2.5× the safe uplift load**. For example: *“Safe (design) uplift capacity = 79 T; Test Load = 197.5 T (2.5 × 79 T)”* <sup>40</sup> <sup>41</sup>. Ensure these values align (197.5 is 2.5× 79 in this case). If multiple piles or multiple tests, specify which pile (e.g., “Initial test pile TP-03”).

**Methodology:** Describe the tension load setup. Uplift tests require a reaction frame or beams anchored (often with kentledge or adjacent piles) to pull the test pile upward. For instance: *“The uplift load was applied using a 500 T capacity hydraulic jack reacting against a frame of girders anchored to the ground”* <sup>42</sup> <sup>43</sup>. *The pile reinforcement was welded to a steel plate or box that the jack pushed against to pull the pile upward* <sup>43</sup>. *Four dial gauges (0.01 mm LC) were positioned at 90° intervals on the pile head to measure uplift movement* <sup>44</sup>, *with bases on independent references 1.5 m away from the pile* <sup>45</sup>. *The jack and gauges were calibrated (certificates provided).”* Explain the loading sequence: increments of ~20% design load, similar to compression. *“The pile was lifted in increments of about 20% of the design load, up to the full test load of 197.5 T, and then unloaded”* <sup>46</sup>. *Each load increment was held for a fixed period (e.g., 5–15 min), and the maximum load was held longer (often 10–60 min or until no further upward movement). Readings of uplift and elastic rebound were taken at each stage.”* Note any specific behavior: e.g., *“Uplift movement was recorded immediately after each load application and after unloading to measure elastic rebound.”* If the test protocol requires a 24-hour hold at max load (less common in uplift, but possibly done to check for creep), mention it accordingly.

**Results:** State acceptance criteria for uplift tests. These often mirror compression criteria: *“The safe uplift capacity is taken as the lesser of: (a) 2/3 of the load at 12 mm upward movement (or 2% of diameter, 12 mm here)”* <sup>47</sup>; *(b) half the load at which the load-displacement curve shows a clear break (signifying failure)”* <sup>48</sup>. The second criterion is specific to uplift (and sometimes lateral), where a **break in the load vs displacement graph** indicates pullout failure – identifying this point may require judgment (and is flagged for LLM assistance). Present the test results similar to compression: *“Maximum uplift at 211.8 T = 9.34 mm, which is less than 12 mm”* <sup>49</sup>. *Total elastic rebound = 3.51 mm; Net uplift (residual displacement) = 5.83 mm* <sup>50</sup>. If a clear break in the load vs uplift curve was observed, note the load at which it occurred. For example, if the graph showed a yield point at, say, 150 T, you would note that *“the load-displacement curve exhibited a break at approximately 150 T, indicating a failure point, so criterion (b) governs in this case.”* In our sample, no such break occurred up to 197.5 T, so criterion (a) governs. Conclude with a statement about capacity: *“Thus, as per the test data and graph, the pile has shown more uplift capacity than the design load of 79 T”* <sup>51</sup>. *The maximum uplift at full load was within the 12 mm limit, with no failure point reached. 79 T is adopted as the safe uplift (tension) capacity for this pile.”* This conclusion follows the pattern of other tests. (If the test failed by reaching a graph break or excessive movement, see Edge Cases below for phrasing.)

**Readings and Graphs:** Provide the uplift test data and graph:

- **Table:** Similar format to vertical load test. Columns: Time, Pressure (kg/cm<sup>2</sup>), Load (T), Dial Gauge readings (1–4), and average uplift. Data will show incremental loading (0, 20%, 40%, etc. of design load) and then unloading steps. After unloading fully, an additional row may show “24 hr rebound” if measured. The agent should compute load from pressure (using jack ram area) if only pressure is given. Also compute net uplift (max minus rebound) if not directly given. Make sure to distinguish uplift direction (movement upward is typically positive in these readings). If the image includes a separate unloading table or notes about rebound after a hold period, include those appropriately (possibly as part of the same table or a short second table).
- **Graph:** Plot Load (T) vs Upward Displacement (mm). Conventionally, upward movement can be plotted on the x-axis and load on y-axis (similar to the compression curve). Clearly label axes (e.g., **“Uplift (mm)”** and **“Load (T)”**). Indicate the 12 mm displacement criterion line and any observed failure/yield point. The graph might show a non-linear curve approaching a plateau if failure occurs. If the test did not reach failure, the curve will end at the maximum load with a displacement below 12 mm. Mark the safe load (usually equal to design load if criteria met) on the graph for clarity. Since

identifying the “break” in the curve (if it exists) may be subjective, consider using the LLM to analyze the curve data for a significant slope change (see AI Assistance section). Annotate accordingly (e.g., a point on the graph labeled “yield point”).

*(Calibration certificate images often accompany uplift tests; handle these as described next if provided.)*

## Common Terminology and Standard Phrases

When generating the narrative portions of the report, use consistent technical terminology as found in real reports. Here are some frequently used terms and phrases, with their meanings and usage:

- **Design Load / Safe Load Capacity:** The working load the pile is designed for. Often, the term “*Safe load capacity*” is used interchangeably with design load in reports. For example: “*Maximum vertical safe capacity of pile = 147 T*”<sup>52</sup> or “*Safe capacity of pile = 550 MT*”<sup>17</sup>. In conclusions, refer to this value: “*...safe vertical load for working piles*”.
- **Test Load:** The maximum load applied during the test, usually a multiple of the design load. E.g., “*The pile was tested to a load of 367.5 T*” (which is  $2.5 \times 147 \text{ T}$ )<sup>1</sup>. Always cross-check this ratio:  $2.5 \times$  for initial tests,  $1.5 \times$  for routine tests, etc., as a validation step.
- **Settlement / Displacement / Deflection:** Use “*settlement*” for vertical compression movement, “*uplift*” or “*uplift displacement*” for upward movement, and “*deflection*” for lateral movement. Always include units (mm). For example, “*total settlement*”, “*net settlement*”, “*lateral deflection*”, “*upward displacement*”.
- **Elastic Rebound:** The amount the pile rebounds (recovers) after unloading. This is measured as the difference between maximum displacement under load and the residual displacement after unloading. Reports often list “*Total Rebound*”<sup>53</sup>. Use this term when presenting results: e.g., “*Total rebound = 2.36 mm*”.
- **Net Settlement/Deflection:** The permanent residual movement after unloading. Calculated as **Net Settlement = Maximum Settlement – Elastic Rebound**<sup>12</sup> (similarly for uplift or lateral). For instance: “*Net Deflection = 1.76 mm*” (2.84 mm max minus 1.08 mm rebound)<sup>36</sup>. This term is commonly used in result summaries.
- **Permissible Limit:** The allowable movement per criteria (e.g., 12 mm or 18 mm for settlement, 5 mm for lateral). Use phrasing like: “*within permissible limits*” or “*did not exceed the permissible deflection of 5 mm*.” In conclusions, it’s common to note: “*...and the deflection was in permissible limits till test load*”<sup>37</sup>.
- **Working Pile vs. Initial Test Pile:** Working piles are those that remain part of the foundation (routine tests), whereas initial test piles may be sacrificial. In reports, initial tests might just call it “*test pile*” and routine tests sometimes explicitly call it “*working pile*.” You may say “*this initial test pile*” or “*the working pile*” appropriately.
- **Load Application and Reaction:** Phrasing like “*load was applied by means of a hydraulic jack of X capacity reacting against a reaction frame*” is standard<sup>54</sup>. Also mention “*pressure gauge and pump*” if relevant. Ensure to specify if multiple jacks or reaction piles were used.
- **Dial Gauge:** Always mention their sensitivity (usually 0.01 mm LC)<sup>55</sup> and placement ( $3 \times$  pile diameter away or on independent frame). E.g., “*Dial gauges (least count 0.01 mm) were fixed on a datum bar 1.5 m away from the pile to measure settlement*”.
- **Observation Timing:** Use the typical time intervals phrasing: “*Readings were taken at 1, 5, 10, 15, 30, and 60 minutes for each load increment*” (modify as per actual intervals used, which might be 1, 10, 20, 30 for lateral<sup>56</sup>, etc.). If the data shows these intervals, list them.

- **Codes and Standards:** Always refer to IS:2911 (Part 4) in methodology since all these tests use it <sup>57</sup>. If IS:14593 (for rock-socket piles) is mentioned in the source material <sup>58</sup>, include it if relevant to project context. For consistency: “conducted as per IS:2911 (Part 4) – 2013” is a safe inclusion in all tests.
- **Safe Load Determination Phrase:** The conclusion should use a standard confident tone, e.g., “can be adopted as the safe load”. We see this exact phrase in examples <sup>14</sup> <sup>21</sup>. Use active voice (“we can say that...”) as needed: e.g., “we can say that the pile has shown more load capacity than design... So, X T can be adopted as safe load.” Avoid passive constructions like “it was determined that X T may be taken...”.
- **Edge-case Terminology:** If a test **fails or doesn’t meet criteria**, do **not** use the “can be adopted as safe load” phrase. Instead, mention the failure: e.g., “the pile did not meet the acceptance criteria, as settlement of 15 mm exceeded the 12 mm limit at only 80% of the target load. The safe load is therefore lower than the design load, and the pile is not acceptable at the design capacity.” Use factual language to describe the outcome. If a **calibration certificate is missing**, explicitly note: “Calibration certificates for the pressure gauge/jack were not available for this test, which may affect the accuracy of load measurements.” This alerts the reader to a data reliability issue. These situations are rare in standard phrasing, but honesty and clarity are paramount in the report.

Using these standardized terms and phrases will ensure the generated reports read naturally and professionally, closely mirroring the style of the provided examples.

## Formulas and Calculations

When extracting data and computing results, apply the standard formulas used in pile load testing. The agent should perform these calculations to fill in report tables and results, marking any interpretation-heavy steps for AI assistance. Key formulas include:

- **Load Calculation from Pressure:** Hydraulic jack readings in the field are often pressure values. Convert to force using:  
**Load (kgf) = Pressure (kg/cm<sup>2</sup>) × Effective Ram Area (cm<sup>2</sup>).**  
 To express in metric tons (T), divide by 1000 (assuming 1 T = 1000 kgf). Thus: **Load (T) = Pressure × Area / 1000**. For example, with a ram area of 2251 cm<sup>2</sup>, 1 kg/cm<sup>2</sup> on the gauge equals 2.251 T of load. The agent should retrieve the **ram area** from the test data (often given in the report header, e.g. “Ram Area = 2251 cm<sup>2</sup>” <sup>59</sup> or listed in methodology) and use it for conversion. *This is a straightforward calculation.* Ensure consistent units (some reports list load directly in metric tons, so conversion may not be needed in those cases).
- **Average Settlement/Deflection:** If multiple dial gauges are used (commonly 2 or 4), compute the average: **Average = (sum of readings) / (number of gauges)**. This smooths out uneven readings. For example, four gauges reading 3.32, 3.99, 3.90, 3.95 mm give an average of ~3.79 mm <sup>60</sup>. Include these averages in the table under an “Average Settlement” column. This calculation is direct arithmetic.
- **Net Settlement / Net Uplift:** After unloading, calculate net settlement (or uplift) as:  
**Net Settlement = Max Settlement – Elastic Rebound** (for compression tests), or  
**Net Uplift = Max Uplift – Elastic Rebound** (for pullout tests).  
 For instance, if max settlement = 9.88 mm and rebound = 2.36 mm, then net settlement = 7.52 mm



<sup>12</sup> . The agent should compute this once the max and rebound values are determined from the data. This formula is straightforward subtraction. It's important for assessing permanent deformation.

- **Safety Criteria Evaluations:** These involve comparing test results to criteria and sometimes require interpolation:

- *Settlement-based capacity:* Identify the load at which the specified settlement limit occurs. Often, the pile does not reach 12 mm or 18 mm exactly, so if the max settlement is below the limit, the criterion (a) isn't controlling (the safe load is then the design load or more). If the limit is exceeded, find the load corresponding to that settlement (linear interpolation between data points if needed) – this step may be complex if data is noisy (**flag such interpolation for [LLM] if uncertainty is high**). Typically, though, if the test is properly done, the limit isn't exceeded for a safe pile.
- *Percentage of diameter criterion:* 2% of diameter is usually similar to the fixed mm values (e.g., 2% of 600 mm = 12 mm). The agent can calculate 2% of pile diameter to double-check the threshold.
- *10% diameter ultimate load criterion:* This is a failure definition (pile is failing if 10% diam settlement reached). Safe load is half of the load at that point. Often not reached in tests, but the code definition is there. If not reached, this criterion isn't governing. If it is reached or exceeded, it implies failure load is around that point. Calculating "half the load at 10% diam settlement" can be done if that point is reached: e.g., 10% of 600 mm = 60 mm; if by extrapolation the pile would carry 300 T at 60 mm, half is 150 T. This is theoretical unless failure actually happened. The agent can note the value but typically criterion (a) will govern safe load in serviceability.
- *Lateral 5 mm criterion:* If the data shows the 5 mm deflection at a certain load, that load is essentially the "failure" load for lateral (serviceability failure). Often, the design load is set to be that failure load divided by a factor (like 2/3 or so), but in our context, we simply check that the design load did not produce >5 mm. If it did, then the safe load would need to be reduced (likely to the load at 5 mm deflection). The agent should find if any reading crosses 5 mm; if none, safe load is just the design load.
- *Uplift break point:* The graph "break" load is where the load-displacement curve flattens, indicating pullout failure. Identifying this precisely might require expert judgment or an AI model to analyze the curve shape. **Mark this step as [LLM]** – i.e., "determine if a clear yield point is present in the load-uplift curve and the load at that point". If found, safe uplift load = half of that load <sup>48</sup> . If not found up to max load, then no failure occurred within test range.
- **Interpretation of Graphs (Break Points):** Any calculation that involves interpreting a graph shape or trend (rather than direct numeric operations) should be flagged for possible AI assistance. For example, determining the exact load at which a curvature change happens (for uplift or lateral) is not a simple formula. The agent should mark such steps with **[LLM]** in the code or workflow to indicate human/AI review. Specifically, "Half the load at which the load-displacement curve shows a clear break" <sup>48</sup> is a criterion that the agent might handle by asking an LLM to analyze the curve data points.

In summary, the agent will handle basic arithmetic (loads, averages, net values) directly, but for any subtle interpretation (e.g., curve yield point, unusual data outliers) it should defer to an AI with a **[LLM]** tag. All calculated values used in the report (like net settlements or safe load recommendations) should be internally cross-checked against criteria to ensure consistency and correctness before being output.

## Data Extraction from Images

To build the report, the agent must reliably extract data from the provided images of tables and graphs. Here's how to approach this extraction:

- **OCR for Tabulated Results:** Use OCR (Optical Character Recognition) to read text from images containing test data tables (e.g., "Record of Pile Load Test" sheets). These tables typically have printed headers and neatly arranged columns, but image quality or scan rotation can be issues. The agent should implement post-processing on OCR output:
- Identify header rows by keywords like "LOAD", "SETTLEMENT", "DEFLECTION", "PRESSURE", "TIME", "Reading 1 2 3 4", etc. <sup>38</sup> <sup>27</sup>. These indicate the structure (e.g., multiple dial gauge readings).
- Normalize header text (OCR might misread "0" vs "O", etc. – correct obvious ones, possibly using an LLM to guess corrections in header labels if needed [LLM]).
- Once columns are determined, parse each subsequent row. Ignore any OCR artifacts like misrecognized lines or combined cells. Heuristics: numeric columns should parse as numbers (possibly with decimals), time might be in hh:mm or mm:ss or minutes.
- **Row grouping:** Ensure that multi-line entries (if any) are consolidated. Often each load step's data might span multiple lines in print (as seen in some PDFs where readings at different minutes are stacked). In the provided examples, readings at 1, 5, 10, etc., minutes were in separate lines below the load step line. The agent should associate these with the correct load step. One approach: use the timestamp or repeated load value to group them. Another approach: some reports list readings in a single row separated by commas or in one cell – handle accordingly.
- **Locate Key Data Points:** The agent should specifically look for:
  - **Initial Zero Readings:** Usually a row with 0 pressure, 0 load, and zero readings at time 0 (or start) <sup>61</sup>. This confirms the initial state – it can be recorded or skipped in output tables (it's often okay to skip the all-zero row in the final report, as it's just the baseline).
  - **Maximum Load Row:** Find the row where the load is highest (should equal the test load). This row is critical for noting max displacement. Also find if there's a 24-hour reading after that (some tables list a reading after many minutes or next day for rebound).
  - **Unloading Data:** After the max load, the table will show decreasing loads. The agent should separate the unloading portion. It might be indicated by a time reset or by notes in "Remarks" column like "unload". If not explicitly marked, detect when the load values start decreasing. Create a separate array for unloading readings for clarity. Also capture final zero load rebound reading.
  - **Rebound/Recovery Measurements:** Sometimes the final unload includes a hold and rebound measure after some time (e.g., measure settlement 5, 10 min after unload). If present, note it as the total rebound.
- **Units and Conversions:** Verify units from the table/headers. Pressure is likely in kg/cm<sup>2</sup>, load in T or kN (the examples show "Load in MT" meaning metric tons) <sup>38</sup>. Settlements in mm. If any values are obviously off (e.g., an OCR mis-read "153.06" as "153.06" – which might be correct – but ensure no mix-up like misplacing decimals or reading ';' as '.'). The agent should correct or flag outliers. For instance, if one gauge reading is wildly different from others (could be an OCR error or actual outlier), consider ignoring or marking it for review.

- **Graph Data Extraction:** If graphs are provided as images, full data digitization is complex. Instead, rely on the corresponding tables for precise numeric values. However, for reporting purposes, the agent might need to extract some info from graphs such as:

- Axis labels and units (to ensure correct labeling in output graph).
- Noted points (sometimes graphs mark safe load or allowable displacement lines with text). OCR the graph image for any text like “12mm” or arrows indicating failure load.
- If no table is available (unlikely in our context, since images of graphs accompany tables), one could attempt to sample points from the graph image, but this is error-prone and beyond typical scope. It’s better to use the table values to regenerate the graph.
- The agent can use an LLM to interpret a graph image if needed, e.g., asking it to estimate where a curve flattens [LLM], but only if absolutely necessary. Usually, stick to numeric table data for conclusions.

- **Heuristics for Table Parsing:** Implement checks like:

- Each row should have the same number of columns as header (if not, OCR might have merged cells incorrectly).
- Pressure and load should generally increase then decrease (monotonic in each phase). If you detect out-of-order values, re-check OCR or assume that marks the transition to unloading.
- Settlement readings generally increase with load (during loading) and decrease on unloading. If a “settlement” value suddenly drops significantly during loading, that could be an OCR error or a note. Verify such anomalies.
- Time stamps normally reset or start new sequence at unload. E.g., in one report, time continued sequentially even during unloading, but in others it might reset to 0 or note a date change. Use context (often the test date might change if overnight). Pay attention to any “Date” column or repeated date entries – that indicates multiple days or a long hold.

- **Unit Conversions:** If the report needs output in different units than given, convert accordingly. The examples stay in metric (T, mm). If the frontend JSON requires a specific unit (say kN), convert T to kN ( $1\text{ T} = 9.81\text{ kN}$ , or often approximated as 10 kN for simplicity if allowed). The guideline here is to maintain consistency: since all sample phrasing uses tons (T) and mm, we will output those by default. Just document clearly in JSON what unit each value is in, or include units in keys (e.g., `"designLoad_T": 147`).

- **Handling Low-Quality Images:** In case of blurry text or scanned handwriting (e.g., sometimes field data might be handwritten or stamped), the OCR might fail on some numbers. If the agent encounters uncertain values:

- It should flag them for [LLM] or manual review.
- The LLM can sometimes infer the correct number from context (for example, if the sequence of loads is known to be 0, 76.53, 114.39, 153.06... and one OCR reading is “153.0G”, the LLM can guess it should be 153.06). Mark these instances and have the LLM attempt a correction, but also ensure to log a warning.

- Use the redundancy in multiple gauge readings to catch errors: e.g., if three out of four gauges show ~2.50 mm and one shows 25.0 mm, it's likely a misplaced decimal OCR error. The agent should correct 25.0 to 2.50 or drop that gauge from averaging (with a note).
- **Image Metadata:** Some images may contain metadata or captions (e.g., a title "Figure 4: Load-Settlement Curve"). Capture such captions if present and associate them with the graph output (for example, as a description). This is less critical, but can enrich the report.

By meticulously extracting the numerical data and relevant text from images, the agent sets the foundation for an accurate report. Any data extraction step that isn't 100% reliable (due to image quality or OCR limits) should be marked for LLM assistance or manual verification.

## Validation Rules and Cross-Checks

Before finalizing a report, the agent should perform several validation checks to ensure the results are consistent and meet expected standards:

- **Design vs Test Load Ratio:** Verify the test was conducted to the correct multiple of the design load. For initial tests, the maximum load should be about 2.5 times the design load <sup>1</sup>; for routine tests, about 1.5 times <sup>2</sup>. If the ratio deviates significantly (e.g., test went only to 2.0× or 2.2× for an initial test), note it in the report (maybe the client stopped early or equipment limit). Similarly, if it overshoot (e.g., 2.6×), mention that the pile was tested slightly beyond requirement. This acts as a sanity check on input data.
- **Correct Diameter and Settlement Limits:** Ensure the permissible settlement/deflection used in criteria matches the pile diameter and test type:
  - For vertical initial tests: 12 mm or 2% of diameter, whichever is smaller <sup>10</sup>. Typically 12 mm governs up to 600 mm dia (since 2% of 600 mm is 12 mm). For larger diameters, 2% might be larger (e.g., 2% of 1200 mm = 24 mm, but code still often uses 12 mm for initial – however, some projects might use 2% rule strictly, in which case initial test on 1200 mm could allow 24 mm; clarify according to IS code: IS 2911-2010 suggests 12 mm for initial regardless of diameter). Our examples used 12 mm for initial 600 mm <sup>10</sup>.
  - For vertical routine tests: 18 mm or 2% of diameter, whichever is less <sup>19</sup>. In the 1200 mm example, 2% is 24 mm, which is greater, so 18 mm governs. If diameter was smaller, 2% might be less than 18 (for ≤900 mm, 2% is ≤18, so 2% would govern).
  - For lateral tests: 5 mm lateral deflection is the usual criterion (for both initial and routine lateral, unless specified otherwise). Check if any project-specific criterion is given (some specs might say 12 mm for lateral as ultimate, but 5 mm is common for serviceability). Our example explicitly uses 5 mm <sup>35</sup>.
  - For uplift tests: 12 mm uplift or 2% diameter (same value in our cases) <sup>47</sup>. 2% of 600 mm = 12 mm, so it matches. If a different diameter, compute 2%. Use the smaller of 12 mm or 2% of dia (some codes don't differentiate initial vs routine for uplift – usually only initial uplifts are done). Also ensure to check if a *break in curve* was observed before 12 mm; if yes, that might govern safe load.

- **Data Consistency:** Cross-check that the reported net settlements/deflections are indeed max minus rebound. The agent should already compute this, but this is to catch any data entry errors. Also verify that the conclusion drawn (pile has more capacity than design load, or not) is consistent with the numbers. For example, if net settlement is slightly above 12 mm at design load, the conclusion should not say “more capacity than design” without clarification. In such a case, the conclusion might need to say the pile did *not* satisfy criteria fully. Ensure no contradiction between “permissible limit” statements and actual values.
- **Graph vs Table Alignment:** If the graph image (if provided) shows a different behavior than the table (e.g., maybe the table has a max settlement of 9.8 mm but the graph might visually show ~10 mm, or a note on the graph says “failure at X T”), reconcile that. Possibly the graph is drawn smoothly and might indicate a yield beyond the raw data – use the data primarily, but mention any graph-indicated values if they add context. For instance, *“the graph suggests a subtle break in slope at ~160 T, though the displacement at that load was under the allowable; no clear failure point was reached.”* This kind of note is only if needed to explain differences.
- **Calibration Certificates Validity:** If a calibration certificate section is present in input, confirm the dates and that the certificate was valid at time of test (usually certificates have a validity period). If the test date is outside the validity or no certificate is given, flag it. In the report, under either Results or an Appendix note, you might include: *“Note: Calibration certificate for the jack pressure gauge is dated [date]. Ensure it was valid during testing.”* Usually, the sample documents include a statement *“The calibration results reported are valid at the time of measurement”* <sup>62</sup> – the agent can reuse such phrasing to assure validity if provided.
- **Creep Criteria (if any):** Some specs require that the final 24-hour settlement (creep) under max load does not exceed a certain value (like 0.1 mm/min in last 30 min). If the data includes 24-hour readings, see if any statement is needed. In many reports, the fact that net settlement is within limits implicitly covers this, but if an explicit creep test was done (24-hour hold), mention something like: *“No significant creep was observed during the 24-hour hold at maximum load (settlement rate < 0.1 mm/hour).”* This is an additional validation if applicable.
- **Conformance to IS Code Requirements:** Ensure the process described meets IS:2911 guidelines:
  - Number of loading cycles (usually just one cycle of loading and unloading, which we have).
  - Duration of each load step (1 hour is per code for routine; initial can be 2.5 hrs at max load for creep, etc.). If any discrepancy (say the data shows shorter holds), mention it neutrally (could be as per client requirement).
  - Unloading sequence being same steps as loading – check table to ensure unload increments roughly match load increments, as required <sup>63</sup>.

Any deviations found by these validations should be explicitly noted in the report output (probably in the Results commentary or a special “Note” section) to maintain transparency. For example, if a test was terminated early, write *“The test was stopped at 80% of the target load due to jack capacity limitations; hence safe load extrapolation is tentative.”* Such honesty is crucial in engineering reports.

## Analysis Tasks Requiring LLM Assistance

Certain parts of the report generation involve interpretation or summarization that goes beyond straightforward rules. We designate these tasks with **[LLM]** to indicate that a Large Language Model (AI) should assist. The developer should build hooks for these tasks:

- **Graph Trendline Identification [LLM]:** Determining qualitative features from the load vs displacement graph, such as the **“clear break” or yield point** in a curve (especially for uplift tests) or confirming that no such break occurred. An LLM can analyze the sequence of load-displacement points to judge if and where the stiffness markedly decreases. For example, it can answer: *“Does the load-settlement curve show a distinct plateau or inflection? If so, at approximately what load?”* Use this for applying criterion like *“half the load at break”* <sup>48</sup>. The LLM's finding would be used to decide which criterion governs the safe load. If the LLM finds no obvious break, the agent assumes criterion (a) (displacement limit) governs.
- **Summarization of Conclusions [LLM]:** While the guidelines provide standard phrasing, an LLM can help tailor the final text based on specific results. For instance, if the pile marginally passed the criteria, the LLM can help phrase that nuance: *“the pile marginally met the criteria with a net settlement of 11.9 mm (just under 12 mm).”* Or if it failed: *“the pile did not satisfy the acceptance criteria, as excessive settlement was observed.”* The agent can feed the raw outcomes (e.g., design load, net settlement, allowable) to the LLM and ask for an appropriate concluding sentence in active voice, matching the tone of examples. This ensures the conclusion section is coherent and context-aware beyond template insertion.
- **Aggregating Multi-Source Readings [LLM]:** In cases where the table parsing is tricky (say OCR splits a row across lines, or readings need correlation), an LLM can assist in restructuring the data. For example, given a chunk of OCR text from a table, the LLM could be prompted to output it as structured JSON or a clean table. This is useful if our algorithmic parsing struggles. Also, for lateral tests, calculating net deflection by subtracting reaction pile movement could be double-checked by the LLM if the approach is unclear (the LLM can be given the raw readings and asked to compute net deflections). Essentially, the LLM can act as a flexible parser/checker for the tabular data.
- **OCR Text Cleanup [LLM]:** Low-quality OCR outputs (e.g., “8.90 mm” instead of “8.90 mm”) can be fixed by context. An LLM, when shown the raw OCR line and perhaps the previous/following lines, can infer and correct errors. This is particularly helpful for misrecognized characters or missing values. The developer can implement a pipeline where after initial OCR, any uncertain tokens (non-numeric where numeric expected, etc.) are sent to an LLM with context for clarification. The corrected output then feeds into the data tables.
- **Natural Language Sections [LLM]:** Sections like Methodology or General can largely follow templates, but if the agent is unsure about phrasing or needs to integrate project-specific info, an LLM can help draft those paragraphs in a coherent way, using the info from input (project name, companies involved, etc.). For example, filling in the client/contractor names in a sentence pulled from the document: *“M/s ZedGeo Systems Pvt Ltd was entrusted with the work of static pile load test...”* <sup>64</sup> can be automated, but an LLM could adjust it for grammar if needed.

Whenever using the LLM, the agent should verify the output for technical accuracy (the LLM might introduce an incorrect value if not carefully constrained). Use LLM suggestions to complement the deterministic calculations, not to replace them, except for interpretive judgments. Mark all such interactions or decision points clearly in code (with comments or tags) for maintainability.

By leveraging AI for these tasks, the agent can handle ambiguous or complex parts of the report generation that are difficult to hard-code, while still following engineering logic and the styles given.

## Output Format: Structured JSON

The final report generated for each test will be serialized as a **JSON object**. This structured output can then be used by a frontend to render the report in a human-readable format (web page, app screen, etc.). The JSON should be organized logically by sections and data, making it easy to format the report. Below is a recommended JSON schema with an example “stub” for each test type, demonstrating the expected structure and some sample content:

### Common Structure:

Regardless of test type, each JSON will include fields for the major sections (general, scope, methodology, results, etc.), as well as nested fields for tables and graphs data. Keys should be consistently named (lowerCamelCase or snake\_case, choose one and stick to it). It's helpful to include the test type as a field too. For instance: `"testType": "IVPLT"`.

### 1. Initial Vertical Load Test (IVPLT) JSON Stub:

```
{
  "testType": "IVPLT",
  "general":
    "Clients decided to carry out a static vertical pile load test on a 600 mm diameter pile to estimate load carrying capacity in vertical direction. M/s ZedGeo Systems Pvt. Ltd., Mumbai was entrusted with this work.",
  "scopeOfWork": {
    "pileDetails": {
      "location": "AGARTAKLI 97 MLD STP",
      "pileDiameter_mm": 600,
      "pileDepth_m": 7.5,
      "concreteGrade": "M25",
      "designLoad_T": 147,
      "testLoad_T": 367.5
    },
    "remarks": "Initial test pile to 2.5× design load as per client requirement."
  },
  "methodology": "The load testing on piles was conducted as per IS: 2911 (Part 4) - 2013. The pile was loaded in increments of ~20% of the design load up to 367.5 T and then unloaded. Load was applied via a 1500 T hydraulic jack reacting against a steel frame. Four dial gauges (0.01 mm LC) recorded settlements at
```

```

each load increment (held for 1 hour each). A 24-hour hold at max load was
conducted to check for creep.",
  "results": {
    "acceptanceCriteria": [
      "Two-thirds of load at 12 mm total settlement or 2% diameter (whichever is
less)",
      "50% of load at 10% pile diameter (settlement = 60 mm)"
    ],
    "maxSettlement_mm": 9.88,
    "elasticRebound_mm": 2.36,
    "netSettlement_mm": 7.52,
    "safeLoadAdopted_T": 147,
    "conclusion": "As per the test data and graph, the test pile showed higher
capacity than the 147 T design load. Therefore, 147 T is adopted as the safe
vertical load for the working piles."
  },
  "readingsAndGraphs": {
    "loadSettlementTable": [
      { "time": "18:45", "pressure_kg_cm2": 15, "load_T": 38.3,
"settlement_readings_mm": [0.36, 0.37, 0.38, 0.35], "avgSettlement_mm": 0.37 },
      { "time": "19:46", "pressure_kg_cm2": 30, "load_T": 76.5,
"settlement_readings_mm": [0.62, 0.81, 0.70, 0.71], "avgSettlement_mm": 0.71 },
      "... additional load increments ...": "...",
      { "time": "23:46", "pressure_kg_cm2": 105, "load_T": 267.9,
"settlement_readings_mm": [??], "avgSettlement_mm": 9.88 }
    ],
    "unloadingTable": [
      { "time": "00:15 (next day)", "pressure_kg_cm2": 0, "load_T": 0,
"settlement_readings_mm": [7.52, 7.50, 7.55, 7.50], "avgSettlement_mm": 7.52,
"remark": "after rebound" }
    ],
    "graph": {
      "title": "Load vs Settlement Curve",
      "xAxis": "Settlement (mm)",
      "yAxis": "Load (T)",
      "dataPoints": [ [0,0], [0.37,38.3], [0.71,76.5], "...", [9.88,382.7] ],
      "safeLoadLine_T": 147,
      "permissibleSettlementLine_mm": 12
    }
  },
  "calibrationCertificate": "Attached (Calibration of jack and gauges dated 01-
Oct-2025)"
}

```

*Explanation:* This JSON contains nested objects for clarity. `scopeOfWork.pileDetails` holds specifics of the pile and loads. The `results` section includes criteria and outcomes (note we used numeric fields for



values like settlement for potential usage in calculations, and a human-readable conclusion string as well). The `readingsAndGraphs` has a detailed table for load vs settlement (truncated with "..." here for brevity) and an unloading table. The graph is described by its axes and key points; a frontend could use `dataPoints` to plot or ignore it if an image is provided instead. The example also shows how to include a calibration note.

## 2. Routine Vertical Load Test (RVPLT) JSON Stub:

```
{
  "testType": "RVPLT",
  "general": "A routine vertical pile load test was conducted on a 1200 mm dia
working pile to verify its load capacity and settlement under 1.5× design
load.",
  "scopeOfWork": {
    "pileDetails": {
      "location": "COOVAM (Chennai Metro EV-03)",
      "pileDiameter_mm": 1200,
      "pileDepth_m": 36.0,
      "concreteGrade": "M35",
      "designLoad_T": 550,
      "testLoad_T": 825
    },
    "remarks": "Routine test pile P357/2 as per IS 2911 to 1.5× design load."
  },
  "methodology": "Load test conducted per IS:2911 (Part 4) – 2013. The pile was
incrementally loaded (≈20% of 550 T each step) up to 825 T using a 1500 T jack.
Each load step was held for 1 hour. Settlement was measured with four dial
gauges (0.01 mm LC). No 24-hr hold was required for this routine test; maximum
load held for 1 hour before unloading.",
  "results": {
    "acceptanceCriteria": [
      "Two-thirds of load at 18 mm settlement or 2% diameter (whichever is less, here
18 mm limit)",
      "50% of load at 10% pile diameter (settlement = 120 mm)"
    ],
    "maxSettlement_mm": 5.79,
    "elasticRebound_mm": 1.92,
    "netSettlement_mm": 3.87,
    "safeLoadAdopted_T": 550,
    "conclusion": "The test pile exhibited only 3.87 mm net settlement at 825 T
(well within the 18 mm limit). It has more capacity than the 550 T design load;
hence, 550 T is adopted as the safe vertical load."
  },
  "readingsAndGraphs": {
    "loadSettlementTable": [
```

```

    { "load_T": 102.0, "settlement_avg_mm": 0.46, "holdDuration_min": 60 },
    { "load_T": 204.1, "settlement_avg_mm": 0.90, "holdDuration_min": 60 },
    "...": "...",
    { "load_T": 841.8, "settlement_avg_mm": 5.05, "holdDuration_min": 60 }
  ],
  "unloadingTable": [
    { "load_T": 0, "rebound_avg_mm": 1.92, "timeAfterUnload_min": 10 }
  ],
  "graph": {
    "title": "Routine Test Load-Settlement",
    "xAxis": "Settlement (mm)",
    "yAxis": "Load (T)",
    "dataPoints": [ [0,0], [0.46,102], [0.90,204], "...", [5.05,842] ],
    "safeLoadLine_T": 550,
    "permissibleSettlementLine_mm": 18
  }
},
"calibrationCertificate": "Attached (Jack & pressure gauge calibrated, valid through Nov-2025)"
}

```

*Note:* In this stub, for brevity, the table is not fully expanded. The structure is similar to IVPLT. We included `holdDuration_min` to explicitly record how long each load was held (optional detail). The results highlight the different settlement limit (18 mm). The conclusion mentions how small the settlement was relative to the limit, which is a nice clarification style.

### 3. Lateral Load Test JSON Stub:

```

{
  "testType": "LateralLoadTest",
  "general": "An initial lateral load test was conducted on a 600 mm dia pile (free-head condition) to determine its safe lateral load capacity with respect to 5 mm deflection criterion.",
  "scopeOfWork": {
    "pileDetails": {
      "pileID": "TP-05",
      "location": "AGARTAKLI 79 MLD STP",
      "pileDiameter_mm": 600,
      "pileDepth_m": 10.31,
      "concreteGrade": "M35",
      "designLateralLoad_T": 3.5,
      "testLateralLoad_T": 8.75
    },
    "remarks": "Initial lateral test to 2.5× design lateral load as per client requirement."
  },
}

```

```

    "methodology": "The test was performed per IS:2911 (Part 4) - 2013. A 50 T hydraulic jack applied lateral load to the pile against a reaction pile. Two dial gauges (0.01 mm LC) measured deflection of the test pile, and two on the reaction to account for any movement. Load increments of ~0.7 T (~20% of 3.5 T) were applied and held until displacement rate <0.1 mm/30min. The final load (8.75 T) was held for 60 min before unloading.",
    "results": {
      "acceptanceCriteria": [
        "Lateral load at 5 mm deflection (safe lateral load should not cause >5 mm displacement)"
      ],
      "maxDeflection_mm": 2.84,
      "elasticRebound_mm": 1.08,
      "netDeflection_mm": 1.76,
      "safeLoadAdopted_T": 3.5,
      "conclusion": "The pile deflected only 2.84 mm at the maximum lateral load of 8.75 T, which is within the 5 mm limit. The pile has demonstrated greater lateral capacity than the 3.5 T design load; therefore, 3.5 T is adopted as the safe lateral load for this pile."
    },
    "readingsAndGraphs": {
      "lateralDeflectionTable": [
        { "load_T": 1.42, "deflection_testPile_mm": 0.30, "deflection_reaction_mm": 0.00, "netDeflection_mm": 0.30, "hold_min": 5 },
        { "load_T": 2.84, "deflection_testPile_mm": 0.76, "deflection_reaction_mm": 0.08, "netDeflection_mm": 0.68, "hold_min": 5 },
        { "load_T": 5.60, "deflection_testPile_mm": 1.40, "deflection_reaction_mm": 0.93, "netDeflection_mm": 0.47, "hold_min": 5 },
        { "load_T": 7.73, "deflection_testPile_mm": 2.61, "deflection_reaction_mm": 1.37, "netDeflection_mm": 1.24, "hold_min": 5 },
        { "load_T": 8.90, "deflection_testPile_mm": 3.24, "deflection_reaction_mm": 2.84, "netDeflection_mm": 0.40, "hold_min": 60 }
      ],
      "graph": {
        "title": "Load vs Lateral Deflection",
        "xAxis": "Deflection (mm)",
        "yAxis": "Lateral Load (T)",
        "dataPoints": [ [0.0,0.0], [0.30,1.42], [0.68,2.84], [1.24,5.60], [1.76,7.73], [2.84,8.75] ],
        "safeLoadLine_T": 3.5,
        "permissibleDeflectionLine_mm": 5
      }
    }
  }
}

```

Notes: The lateral table combines test and reaction deflection and gives net (test minus reaction). We listed some example data points to illustrate (not actual, just indicative). The conclusion shows how to mention the deflection and criterion. The graph points show deflection on x-axis.

#### 4. Uplift/Pullout Test JSON Stub:

```
{
  "testType": "UpliftLoadTest",
  "general": "An initial uplift (tension) pile load test was carried out on a 600 mm dia pile to determine its safe uplift capacity.",
  "scopeOfWork": {
    "pileDetails": {
      "pileID": "TP-03",
      "location": "AGARTAKLI 97 MLD STP",
      "pileDiameter_mm": 600,
      "pileDepth_m": 8.5,
      "concreteGrade": "M25",
      "designUpliftLoad_T": 79,
      "testUpliftLoad_T": 197.5
    },
    "remarks": "Initial pullout test to 2.5× design uplift load."
  },
  "methodology": "Testing was done as per IS:2911 (Part 4) – 2013 for uplift. The pile reinforcement was welded to a loading frame and a 500 T hydraulic jack applied upward load. Increments of ~15.8 T (20% of 79 T) were applied up to 197.5 T. Four dial gauges (0.01 mm) recorded uplift at the pile head. Each increment was held ~5–10 min, and the max load was held for 15 min. The rebound was measured after unloading to determine net uplift.",
  "results": {
    "acceptanceCriteria": [
      "Two-thirds of load at 12 mm uplift or 2% diameter (whichever is less, here 12 mm)",
      "Half the load at which the load-uplift curve shows a clear break (yield point)"
    ],
    "maxUplift_mm": 8.46,
    "elasticRebound_mm": 3.75,
    "netUplift_mm": 4.71,
    "safeLoadAdopted_T": 79,
    "conclusion": "The pile showed 8.46 mm uplift at 162.4 T (below the 12 mm criterion) and no distinct failure point up to 197.5 T. Thus, it exceeded the design uplift load of 79 T without reaching the allowable displacement. 79 T is taken as the safe uplift capacity for the pile."
  },
  "readingsAndGraphs": {
    "upliftTable": [
      { "load_T": 21.2, "uplift_readings_mm": [0.10, 0.12, 0.15, 0.11],
```

```

"avgUplift_mm": 0.12 },
  { "load_T": 42.4, "uplift_readings_mm": [0.37, 0.38, 0.49, 0.52],
"avgUplift_mm": 0.44 },
  { "load_T": 63.5, "uplift_readings_mm": [0.58, 0.65, 0.99, 1.00],
"avgUplift_mm": 0.81 },
  { "load_T": 84.7, "uplift_readings_mm": [1.88, 1.70, 2.09, 2.10],
"avgUplift_mm": 1.94 },
  { "load_T": 105.9, "uplift_readings_mm": [2.78, 2.80, 3.15, 3.40],
"avgUplift_mm": 3.03 },
  { "load_T": 127.1, "uplift_readings_mm": [3.84, 3.87, 4.80, 4.50],
"avgUplift_mm": 4.25 },
  { "load_T": 150.0, "uplift_readings_mm": [5.10, 5.00, 6.20, 5.80],
"avgUplift_mm": 5.53 },
  { "load_T": 162.4, "uplift_readings_mm": [7.90, 8.10, 8.60, 8.25],
"avgUplift_mm": 8.46 },
  { "load_T": 0, "rebound_readings_mm": [4.70, 4.72, 4.68, 4.75],
"avgRebound_mm": 4.71, "remark": "Net residual uplift after unload" }
],
"graph": {
  "title": "Load vs Uplift Displacement",
  "xAxis": "Upward Displacement (mm)",
  "yAxis": "Load (T)",
  "dataPoints": [ [0,0], [0.12,21.2], [0.44,42.4], [0.81,63.5], [1.94,
84.7], [3.03,105.9], [4.25,127.1], [5.53,150.0], [8.46,162.4] ],
  "safeLoadLine_T": 79,
  "permissibleUpliftLine_mm": 12
}
},
"calibrationCertificate": "Not provided"
}

```

*Notes:* We filled in more of the table to illustrate the progression (the values are hypothetical but consistent). We also showed an example where no calibration certificate was provided. The conclusion here explicitly notes no failure point and that it met criteria. If a yield had occurred, we'd adjust accordingly (e.g., if a break was at 150 T, then safe load would be 75 T per that criterion, which is below design 79 T, implying design was too high – that would be a fail case to describe).

Each JSON stub above demonstrates how to structure the data and text. In practice, the agent will populate all fields with actual values from the image inputs and computed results. The JSON format ensures that frontend code can easily extract each part (for example, display `general` as a paragraph, iterate `acceptanceCriteria` as bullet points, render tables from the arrays, plot graphs from `dataPoints`, etc.).

Finally, ensure the JSON is valid (no trailing commas, proper quotations, etc.). Including sample stubs like above (perhaps with some placeholder “...” for long arrays) in documentation helps consumers understand how to parse and use the output.

## Writing Style and Tone Guidelines

When composing the textual sections of the report (which the agent will output via templates or LLM), follow these style requirements to maintain clarity and professionalism:

- **Use Active Voice:** Favor direct, active constructions over passive. For example, write *"The pile achieved 7.52 mm net settlement under 2.5× load"* instead of *"A net settlement of 7.52 mm was achieved by the pile."* Active voice makes the report more readable and confident. In conclusions, always explicitly state the actor or subject (often "the pile") and the outcome. E.g., *"the test pile has shown more load carrying capacity..."* <sup>13</sup> is preferred over *"it is shown that the pile has..."*.
- **Maintain a Technical Yet Accessible Tone:** The language should be formal and technical (using correct engineering terms as listed in the terminology section), but not overly verbose or filled with unnecessary jargon. Assume the reader has basic understanding of pile tests but still explain key points (especially criteria and outcomes) clearly. For instance, it's good to say *"which is within the permissible limit"* or *"exceeded the allowable settlement"* to tie numbers to acceptance criteria, rather than just stating numbers. Avoid slang or conversational tone; remain professional and objective.
- **Keep Paragraphs Focused and Concise:** Aim for paragraphs of about 3-5 sentences. Each paragraph should cover a single topic or result. In this guideline document, for example, each section's description is broken into digestible chunks. The same should apply in the generated report – e.g., one short paragraph for methodology, one for results discussion, one for conclusion. This prevents overwhelming the reader with wall-of-text sections.
- **Bullet Points for Lists:** Use bulleted lists for clarity when enumerating:
  - Acceptance criteria (as we have done in results sections) should be in bullet form for easy reading <sup>10</sup>.
  - If listing multiple codes or references, bullets or comma-separated lists are fine (the sample reports bullet the codes <sup>4</sup>).
  - Use numbering only if there is a stepwise sequence (for example, if describing step-by-step loading procedure, one could use an ordered list, but narrative form is usually sufficient for methodology).
- **Consistent Section Headings:** The report sections should appear with their numbering (1.0, 2.0, etc.) and titles as given (General, Scope of Work, Methodology, Results, etc.). In the JSON, we don't include numbering (the frontend can add if needed), but in the rendered output, these should be clearly marked. Also, sub-sections like 2.1 Pile Details, 2.2 Test Load, or 4.1 Acceptance Criteria, can be included to mirror the format of the sample reports <sup>65</sup> <sup>66</sup>. For example, under Results, you might boldface *"4.1 Acceptance Criteria..."* followed by the bullet list, then another subheading for *"4.2 Test Observations"* for the measured results. Maintaining this structure makes the report familiar to industry readers.
- **No Unnecessary Padding:** Keep the content relevant to the test. Do not add extraneous information not provided by the data. For instance, don't speculate or add design recommendations beyond the scope. The tone should remain factual: report what was done, what was observed, and what is concluded from it.

- **Referencing Data in Text:** When writing the narrative, reference the key data points plainly. For example, “*maximum settlement of 9.88 mm was recorded at 382.65 T*” <sup>12</sup> or “*the pile deflected 2.84 mm at the maximum load of 8.75 T*”. Including the numbers in the sentences (as well as in tables) reinforces the conclusions. However, avoid overloading sentences with too many numbers; use a table for detailed series of readings and use text to highlight just the crucial ones (max, net, criteria).
- **Edge Case Wording:** If the test had anomalies (e.g., equipment malfunction, test stopped early, etc.), clearly state them in neutral tone. For example, “*Note: Due to jack capacity limitations, the test could not reach the full target load; results must be interpreted with caution.*” Similarly, for missing calibration: “*No calibration certificate was available for the pressure gauge, so slight error in load values is possible.*” Such notes should typically be in the Results section or footnotes, not buried in methodology. They should stand out either as italicized notes or a separate paragraph.

By adhering to these style guidelines, the generated reports will remain clear, professional, and easy to read. This ensures that even though the content is produced automatically, it meets the quality expected by engineers and project stakeholders.

**Conclusion:** Following the above guidelines, the coding agent can reliably convert raw image-based test data into comprehensive pile load test reports. Each report will include all necessary technical details and standardized language drawn from real examples, ensuring consistency. The use of structured JSON output will facilitate easy rendering on the frontend, and the incorporation of validation checks and LLM assistance points will enhance accuracy and readability. By implementing these instructions, the agent will produce outputs that closely mimic expert-written reports, enabling quick generation of results for each pile test performed.

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1 3 4 5 6 10 11 12 13 14 15 52 59 60 62 Report IVPLT TP-04 600mm (147T)..pdf  
file:///file-YAwZxYoYPB1EDYr3tp27ig

2 7 8 9 16 17 18 19 20 21 54 55 57 64 RVPLT (P357-2) COOVAM 550T (1.2m) Report..pdf  
file:///file-LvJVrzS9K6UYgbhYQq44Av

22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 53 56 61 63 65 66 NASIK initial Lateral  
Report 8.75 AGARTAKLI.pdf  
file:///file-SJcyDVsyLVb5TSTEW16EPz

39 40 41 42 43 44 45 46 47 48 49 50 58 TP-03 Pullout load test report.pdf  
file:///file-ERGtVFtMSiU1QaTmjEP3YS

51 TP-01 Pullout load test report..pdf  
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